Analysis Tool: Causal Loop Diagram

What Are Causal Loop Diagrams (CLD)?

- Problem analysis tool that:
  - Allows you to consider the possibility of feedback loops contributing to the presence of the issue
  - Helps you to understand the underlying root **causes** behind problems
  - A helpful representation of the interconnected nature of systems that take place in the city
- Made up of 2 components
  - Boxes → These represent the variables, or the things that can change
  - Connections → These represent the causal connections between variables i.e. where one variable changing causes another to change

CLDs in CDE2501

Please do not be scared off by this tool!

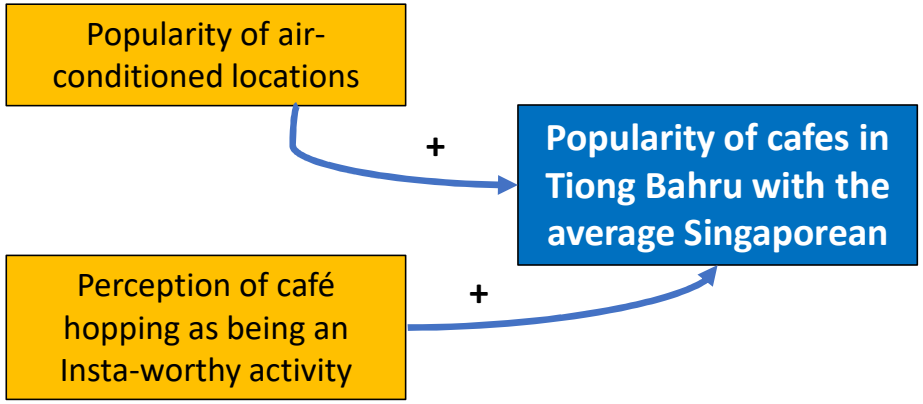
- This is just a tool to help you to understand the underlying issues behind the problem you have identified
- It is NOT a compulsory requirement for your presentations
  - However, if you would like to show it as part of your presentation and it helps you to explain your case, then by all means include it
  - That being said, you are encouraged to use this tool in your thinking to understand the problem as a whole

Issue of focus: Gentrification in Tiong Bahru

Popularity of cafes in Tiong Bahru with the average Singaporean

Step 1: Identify a variable that is closely connected to the issue you are investigating

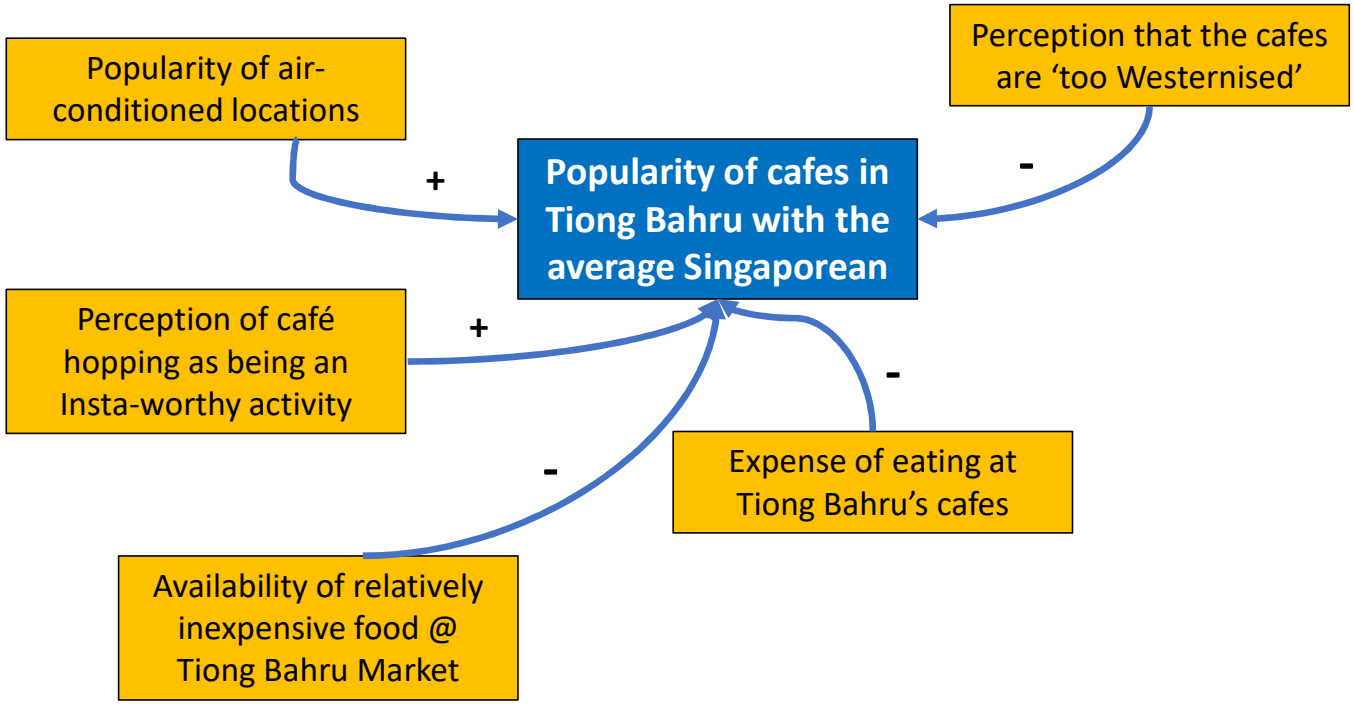
Issue of focus: Gentrification in Tiong Bahru



Step 2: Identify some immediate causes of this central variable that can be labelled with a '+'

'+' represents how when one variable increases, the other will increase as well OR when one variable decreases, the other will decrease as well

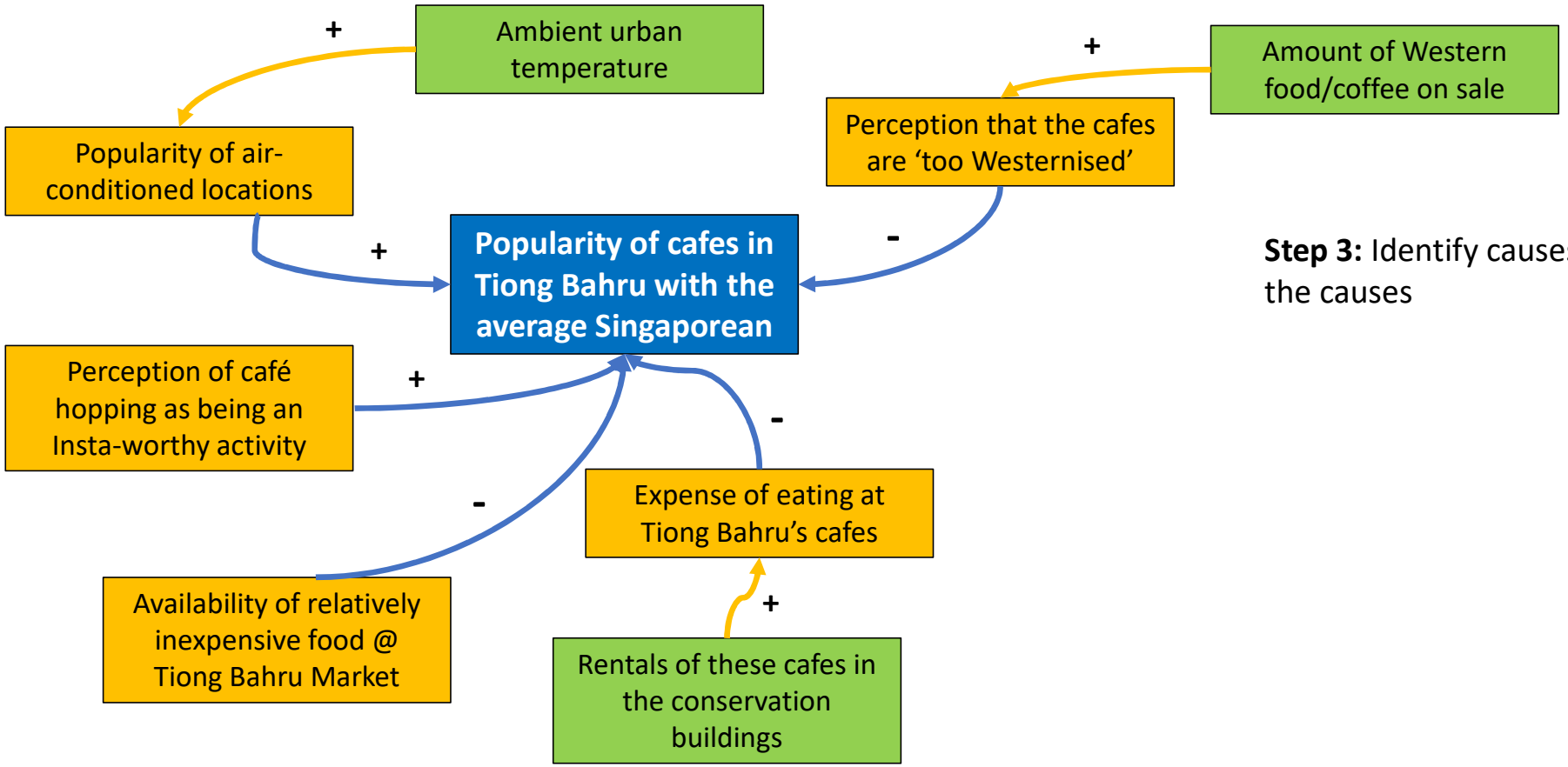
Issue of focus: Gentrification in Tiong Bahru



Step 2: Identify some immediate causes of this central variable that can be labelled with a '-'

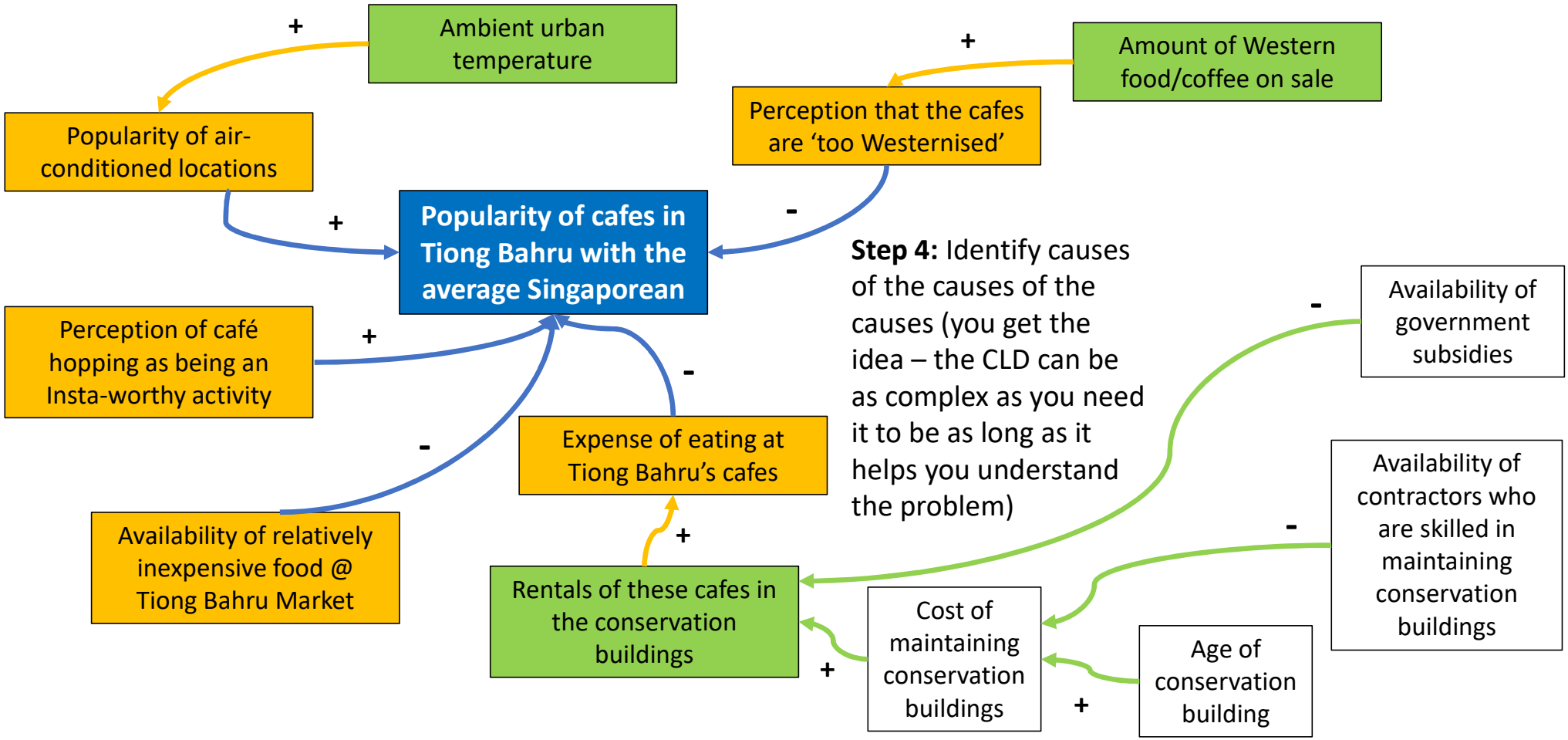
'-' represents how when one variable increases, the other will **decrease** OR when one variable decreases, the other will **increase**

Issue of focus: Gentrification in Tiong Bahru



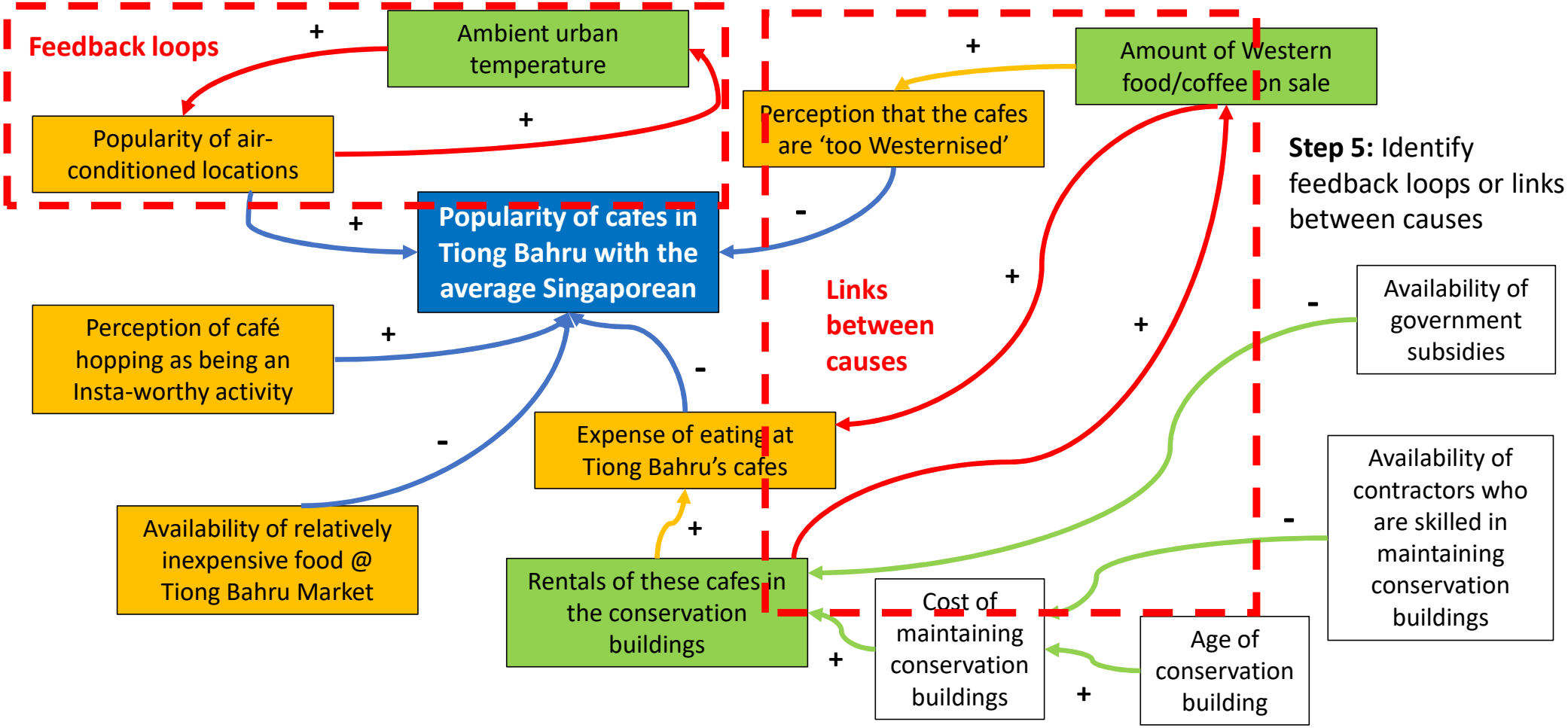
Step 3: Identify causes of the causes

Issue of focus: Gentrification in Tiong Bahru



Step 4: Identify causes of the causes of the causes (you get the idea – the CLD can be as complex as you need it to be as long as it helps you understand the problem)

Issue of focus: Gentrification in Tiong Bahru



Step 5: Identify feedback loops or links between causes

Why are CLDs Helpful?

- Helps you to understand root causes behind the problem you have identified
- In turn, this helps you to craft your solutions to the problem (deliverable in final presentation)
  - Addressing the root causes is normally more durable that addressing the surface problem only
  - You can also identify feedback loops that can either be removed or enhanced to attempt to solve the problem